

Unit 6

ELECTROCHEMISTRY

24 items · 43 marks

Syllabus sections in this unit

6.1 Redox processes: electron transfer and changes in oxidation number (oxidation state)

Unit 6: Electrochemistry

1. M/J 26/P21/Q3(d)

Context

- 3 Separate samples of NaCl, NaBr and NaI each react with concentrated H₂SO₄. Table 3.1 shows the observations for each reaction.

Table 3.1

reaction	compound	observations
1	NaCl	- effervescence - steamy fumes that turn blue litmus paper red
2	NaBr	- effervescence - steamy fumes initially that turn blue litmus paper red - brown vapour - pungent smell of SO₂
3	NaI	- effervescence - steamy fumes initially that turn blue litmus paper red - purple vapour - pungent smell of SO₂ and then a rotten egg smell of H₂S

- (d) Complete Table 3.2.

Table 3.2

species	oxidation number of sulfur
H ₂ SO ₄	
SO ₂	
H ₂ S	

[2]

2. M/J 26/P21/Q4(a)(ii)

Context: part (a) (introduction)

- 4 (a) Compound **Z** is oxidised to produce organic compound **W**, C₅H₆O₇, by heating under reflux with an excess of acidified Cr₂O₇²⁻(aq).

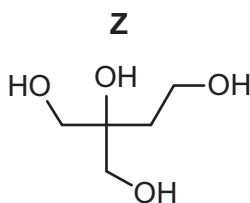
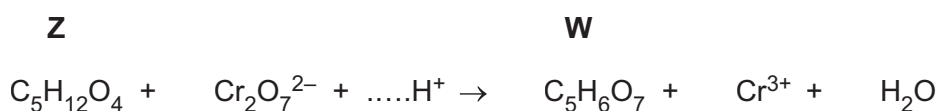


Figure 4.1

- (ii) The complete oxidation of 0.025 mol of **Z** requires 0.050 mol of Cr₂O₇²⁻(aq). **W** is produced in this reaction.

Balance the equation for this reaction.

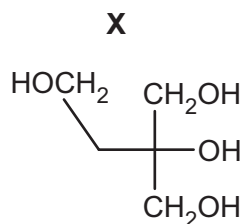


[1]

3. M/J 26/P22/Q4(c)

also Unit 17 (2 marks); this unit 2 marks

Context: part (b) (introduction)

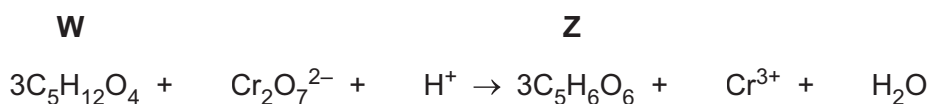
(b) The structure of compound **X** is shown in Figure 4.2.**Figure 4.2**

The molecular formula of **X** is $\text{C}_5\text{H}_{12}\text{O}_4$. Complete oxidation of 0.255 g of **X** occurs when it reacts with exactly 10.0 cm^3 of $0.500 \text{ mol dm}^{-3}$ acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$.

(c) (i) Compounds **X** and **W** are a pair of structural isomers.

0.0012 mol of **W** is completely oxidised when it is heated under reflux with exactly 0.0020 mol of acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$ to produce compound **Z**.

Complete the equation for this reaction.



[2]

(ii) Each molecule of **W** contains four hydroxyl groups. Only two of these are primary hydroxyl groups.

Table 4.1 shows the observations when separate samples of **Z** are added to each reagent.

Table 4.1

reagent	observations with Z
2,4-dinitrophenylhydrazine (2,4-DNPH)	orange precipitate
Fehling's reagent	none

Use the information in **4(c)** to suggest the structure of **W**.

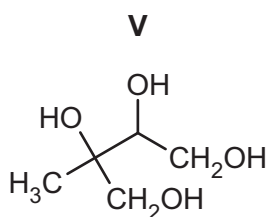


[2]

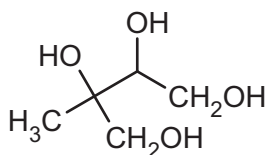
4. M/J 26/P23/Q4(d)(ii)

Context

- 4** The structure of compound **V** is shown in Figure 4.1.

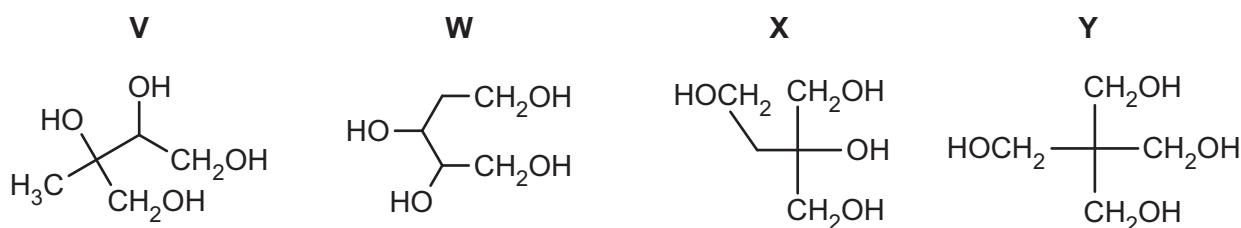
**Figure 4.1**

Context: Fig. 4.1

**Figure 4.1**

Context: part (c) (introduction)

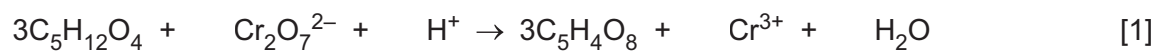
- (c)** Compounds **W**, **X** and **Y** are structural isomers of **V**.

**Figure 4.2**

Context: part (d) (introduction)

- (d)** Exactly 0.03 mol of one of the isomers, **V**, **W**, **X** or **Y**, is completely oxidised by 0.08 mol of acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$ to produce only **one** organic compound with molecular formula $\text{C}_5\text{H}_4\text{O}_8$.

- (ii)** Complete the equation for this reaction.



5. M/J 25/P21/Q4(b)(iii)

Context

4 The structure of vitamin C is shown in Fig. 4.1.

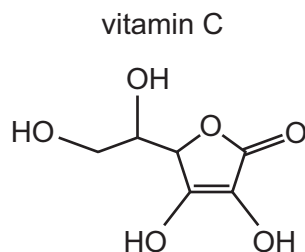


Fig. 4.1

Context: part (b) (introduction)

(b) The concentration of vitamin C is found by titration with $\text{I}_2(\text{aq})$.

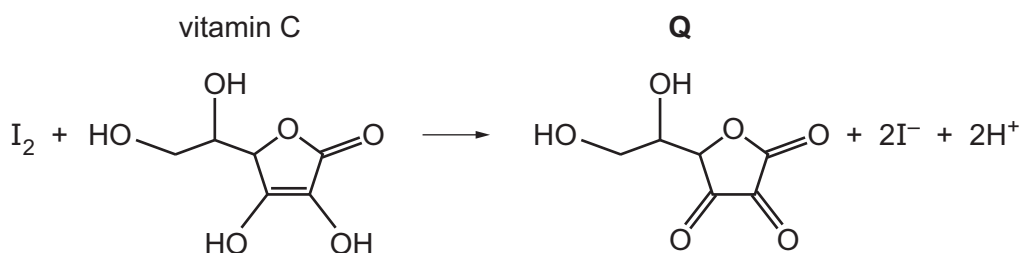


Fig. 4.2

A vitamin C tablet is dissolved in water to produce 200.0 cm³ of vitamin C solution. 5.00 cm³ of this vitamin C solution is added to a flask with approximately 150 cm³ of water and an indicator. Exactly 28.40 cm³ of 5.00 × 10⁻⁴ mol dm⁻³ I₂(aq) reacts with the sample of vitamin C solution in the flask.

[M_r : vitamin C, 176]

(iii) Deduce the role of $\text{I}_2(\text{aq})$ in the reaction in Fig. 4.2.

[1]

6. M/J 25/P24/Q2(b)(ii)

Context: part (b) (introduction)

(b) SCl_2 is a red liquid that reacts with water.



(ii) When SCl_2 reacts with water, the SCl_2 is broken down and a disproportionation reaction occurs.

State the oxidation numbers of sulfur in SCl_2 and H_2SO_3 .

oxidation number of S in SCl_2

oxidation number of S in H_2SO_3

[2]

7. MAR 25/P22/Q1(a)

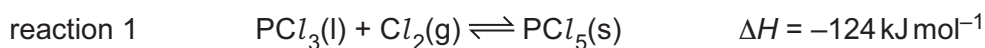
also Unit 7 (2 marks), Unit 2 (2 marks), Unit 4 (2 marks); this unit 2 marks

Context

1 Phosphorus and chlorine are elements in Period 3 of the Periodic Table.**(a)** Chlorine forms three different compounds with phosphorus.The most common compounds are PCl_3 and PCl_5 .**(i)** Complete Table 1.1.**Table 1.1**

compound	oxidation number of P	oxidation number of Cl
PCl_3		
PCl_5		

[2]

(ii) In a closed system, PCl_3 and PCl_5 exist in an equilibrium mixture as shown in reaction 1.Deduce **two** conditions that favour the production of PCl_5 in reaction 1.

1

2

[2]

(iii) The third compound of phosphorus and chlorine, **W**, has a relative molecular mass, M_r , between that of PCl_3 and PCl_5 . The compound contains 69.6% by mass of chlorine.Determine the molecular formula of **W**.molecular formula of **W** = [2]**(iv)** **W** is a liquid at room temperature and pressure. It reacts vigorously with water to form an acidic solution.Suggest the structure and bonding in **W**. Explain your answer.

[2]

8. O/N 24/P21/Q2(c)(ii)-(iii)

Context

2 Chlorine is one of the elements in Group 17 of the Periodic Table.

Context: part (c) (introduction)

(c) CHCl_3 is another halogenoalkane. CHCl_3 forms when propanone reacts with NaClO . NaClO is made from chlorine in a disproportionation reaction.**(ii)** Define disproportionation.

[1]

(iii) Write numbers in the boxes to balance the equation showing the reaction of propanone with NaClO .

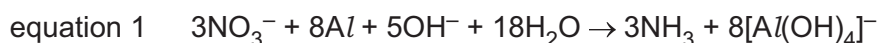
[1]

9. O/N 24/P22/Q3(b)(i)-(ii)

Context

3 Nitrogen and phosphorus are elements in Group 15 of the Periodic Table.

Context: part (b) (introduction)

(b) A common test for nitrates is the reaction with NaOH and Al . Equation 1 shows the reaction.**(i)** Deduce the oxidation state of nitrogen in NO_3^- .

[1]

(ii) Identify the species that is oxidised in equation 1.

[1]

10. O/N 24/P23/Q2(c)(ii)-(iii)

Context

2 Chlorine is one of the elements in Group 17 of the Periodic Table.

Context: part (c) (introduction)

(c) CHCl_3 is another halogenoalkane. CHCl_3 forms when propanone reacts with NaClO . NaClO is made from chlorine in a disproportionation reaction.**(ii)** Define disproportionation.

[1]

(iii) Write numbers in the boxes to balance the equation showing the reaction of propanone with NaClO .

[1]

11. M/J 24/P23/Q1(d)(i)

Context: part (d) (introduction)

- (d) The oxide of silicon reacts with calcium oxide in an addition reaction to produce calcium silicate, CaSiO_3 . The oxidation number of calcium in CaSiO_3 is +II.

- (i) Deduce the oxidation number of silicon in calcium silicate.

oxidation number of silicon

[1]

12. MAR 24/P22/Q1(b)(i)

Context

- 1 Bismuth is an element in Group 15 of the Periodic Table.

Context: part (b) (introduction)

- (b) Bismuth reduces water to form bismuth oxide, Bi_2O_3 . A colourless gas that ignites with a squeaky pop also forms.

- (i) Construct an equation for the reduction of water by bismuth.

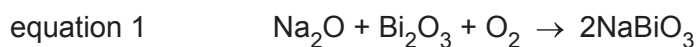
[1]

13. MAR 24/P22/Q1(c)

Context

- 1 Bismuth is an element in Group 15 of the Periodic Table.

- (c) Bi_2O_3 can be used to form NaBiO_3 , as shown in equation 1.



- (i) Deduce the oxidation number of Bi in Bi_2O_3 and in NaBiO_3 .

oxidation number of Bi:

in Bi_2O_3 in NaBiO_3

[1]

- (ii) Identify the reducing agent in equation 1.

[1]

14. MAR 24/P22/Q1(d)(i)

Context

- 1** Bismuth is an element in Group 15 of the Periodic Table.

Context: part (d) (introduction)

- (d)** NaBiO_3 is an oxidising agent with similar properties to KMnO_4 .

Fig. 1.1 shows an example of the use of NaBiO_3 as an oxidising agent.

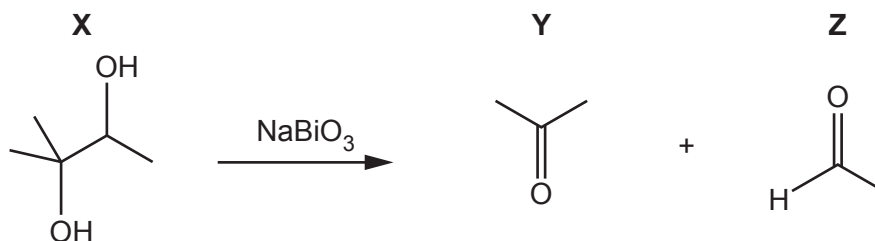


Fig. 1.1

- (i)** Explain the term oxidising agent.

[1]

15. O/N 23/P21/Q3(a)

Context

- 3** Potassium chlorate, KClO_3 , is widely used as an oxidising agent and to make $\text{O}_2(\text{g})$.

- (a)** Define oxidising agent.

[1]

16. O/N 23/P21/Q3(c)

Context

- 3** Potassium chlorate, KClO_3 , is widely used as an oxidising agent and to make $\text{O}_2(\text{g})$.

- (c)** When KClO_3 is heated without a catalyst, KClO_4 and KCl form.



Explain why this reaction is described as a disproportionation reaction.

[1]

17. O/N 23/P22/Q3(b)(iii)

Context

3 Phosphoric(V) acid, H_3PO_4 , is used in both inorganic and organic reactions.

Context: part (b) (introduction)

(b) H_3PO_4 is also formed in the process shown in reaction 1.

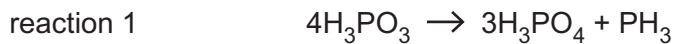


Table 3.1 shows some relevant thermodynamic data.

Table 3.1

compound	enthalpy change of formation, $\Delta H_f / \text{kJ mol}^{-1}$
H_3PO_3	−972
H_3PO_4	−1281
PH_3	+9

(iii) Explain why reaction 1 is a disproportionation reaction.

Explain your reasoning with reference to relevant oxidation numbers.

[2]

18. O/N 23/P23/Q3(a)

Context

3 Potassium chlorate, KClO_3 , is widely used as an oxidising agent and to make $\text{O}_2(\text{g})$.

(a) Define oxidising agent.

[1]

19. O/N 23/P23/Q3(c)

Context

3 Potassium chlorate, KClO_3 , is widely used as an oxidising agent and to make $\text{O}_2(\text{g})$.

(c) When KClO_3 is heated without a catalyst, KClO_4 and KCl form.



Explain why this reaction is described as a disproportionation reaction.

[1]

20. M/J 23/P22/Q2(e)(ii)

Context

2 Chlorine is a reactive element. It forms many compounds.

Context: part (e) (introduction)

(e) Bleach is used as a cleaning product to kill bacteria. It is made by adding compounds like sodium chlorate(I), NaClO , to water.

(ii) Sodium chlorate(I), NaClO , reacts with hydrogen peroxide to produce sodium chloride, water and oxygen gas.

Construct an equation for this reaction.

[1]

21. M/J 23/P23/Q1(d)(i)-(ii)

Context

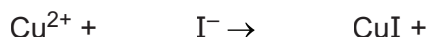
1 Copper is used in electrical equipment. It has a melting point of 1085°C .

Context: part (d) (introduction)

(d) When KI(aq) is added to $\text{CuSO}_4\text{(aq)}$ the blue-coloured solution turns brown and a white precipitate of CuI(s) is seen.

The reaction between copper ions and iodide forms only two products.

(i) Complete the equation for this reaction.



[1]

(ii) Identify the oxidising agent in this reaction. Explain your answer in terms of electron transfer.

[1]

22. M/J 22/P22/Q1(b)(i)-(ii)

Context: part (b) (introduction)

(b) When magnesium is heated in air, magnesium oxide, MgO , is the major product. Smaller amounts of magnesium nitride, Mg_3N_2 , are also made.

(i) Calculate the oxidation number for magnesium and for the nitrogen species in Mg_3N_2 to complete Table 1.1.

Table 1.1

species	magnesium in Mg_3N_2	nitrogen in Mg_3N_2
oxidation number		

[1]

(ii) Identify the type of reaction which takes place between magnesium and nitrogen. Explain your answer.

[1]

23. MAR 22/P22/Q2(a)(i)

Context

2 Some oxides of elements in Period 3 are shown.



Context: part (a) (introduction)

(a) Na reacts with O_2 to form Na_2O . Na is the reducing agent in this reaction.

(i) Define reducing agent.

[1]

24. MAR 22/P22/Q2(c)(ii)

Context

2 Some oxides of elements in Period 3 are shown.



Context: part (c) (introduction)

(c) P_4O_6 is a white solid that has a melting point of 24°C . Solid P_4O_6 reacts with water to form H_3PO_3 .

(ii) Determine the oxidation number of P in H_3PO_3 .

[1]

Answers Section

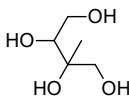
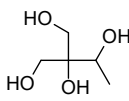
1. M/J 26/P21/Q3(d)

3(d)	species	oxidation number of sulfur	2 •✓✓ A 6 (+) 4 (+) and 2– A VI (+) AND IV (+) AND II – R S ⁺⁴ S ⁺⁶ or S ^{–2} R S ⁴⁺ S ⁶⁺ or S ^{2–}
	H ₂ SO ₄	(+)VI OR (+)6	
	SO ₂	(+)IV OR (+)4	
	H ₂ S	–II OR –2	

2. M/J 26/P21/Q4(a)(ii)

4(a)(ii)	$\text{C}_5\text{H}_{12}\text{O}_4 + 2\text{Cr}_2\text{O}_7^{2-} + 16\text{H}^+ \rightarrow \text{C}_5\text{H}_6\text{O}_7 + 4\text{Cr}^{3+} + 11\text{H}_2\text{O}$	1	
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3. M/J 26/P22/Q4(c)

4(c)(i)	$3\text{C}_5\text{H}_{12}\text{O}_4 + 5\text{Cr}_2\text{O}_7^{2-} + 40\text{H}^+ \rightarrow 3\text{C}_5\text{H}_6\text{O}_8 + 10\text{Cr}^{3+} + 29\text{H}_2\text{O}$ M1 $5\text{Cr}_2\text{O}_7^{2-}$ M2 fully correct equation	2	
4(c)(ii)	HOCH ₂ CH(OH)C(OH)(CH ₃)CH ₂ OH OR (HOCH ₂) ₂ C(OH)(CHOHCH ₃)  OR  - molecular formula C₅H₁₂O₄ 2 primary alcohol groups 1 secondary alcohol group 1 tertiary alcohol group •✓•✓	2	A unambiguous structure I any name attempted Max 1 all correct but 1 or more connectivity issue e.g. C–H–O

4. M/J 26/P23/Q4(d)(ii)

4(d)(ii)	$3\text{C}_5\text{H}_{12}\text{O}_4 + 8\text{Cr}_2\text{O}_7^{2-} + 64\text{H}^+ \rightarrow 3\text{C}_5\text{H}_4\text{O}_8 + 16\text{Cr}^{3+} + 44\text{H}_2\text{O}$	1	
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5. M/J 25/P21/Q4(b)(iii)

4(b)(iii)	oxidising agent	1
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6. M/J 25/P24/Q2(b)(ii)

2(b)(ii)	oxidation number of S in $\text{SCl}_2 = (+)2 / (+)\text{II}$ oxidation number of S in $\text{H}_2\text{SO}_3 = (+)4 / (+)\text{IV}$	2
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7. MAR 25/P22/Q1(a)

1(a)(i)	• (+)3 / III • –1 / I • (+)5 / V • –1 / I	2
1(a)(ii)	high pressure low temperature	2
1(a)(iii)	ratio of moles of P : Cl = $\frac{30.4}{31.0} : \frac{69.6}{35.5} = 0.98 : 1.96 (= 1 : 2)$ molecular formula = P_2Cl_4	2
1(a)(iv)	(structure =) simple / molecular because low melting point / weak intermolecular forces (bonding =) covalent because hydrolysed	2

8. O/N 24/P21/Q2(c)(ii)-(iii)

2(c)(ii)	a reaction where a species / same species is both oxidised and reduced simultaneously / in the same reaction	1
2(c)(iii)	$\text{CH}_3\text{COCH}_3 + 3\text{NaClO} \rightarrow 1\text{CHCl}_3 + 1\text{CH}_3\text{COONa} + 2\text{NaOH}$	1

9. O/N 24/P22/Q3(b)(i)-(ii)

3(b)(i)	(+)5 / V	1
3(b)(ii)	aluminium / Al	1

10. O/N 24/P23/Q2(c)(ii)-(iii)

2(c)(ii)	a reaction where a species / same species is both oxidised and reduced simultaneously / in the same reaction	1
2(c)(iii)	$\text{CH}_3\text{COCH}_3 + 3\text{NaClO} \rightarrow 1\text{CHCl}_3 + 1\text{CH}_3\text{COONa} + 2\text{NaOH}$	1

11. M/J 24/P23/Q1(d)(i)

1(d)(i)	(+)IV	1
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12. MAR 24/P22/Q1(b)(i)

1(b)(i)	$2\text{Bi} + 3\text{H}_2\text{O} \rightarrow \text{Bi}_2\text{O}_3 + 3\text{H}_2$	1
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13. MAR 24/P22/Q1(c)

1(c)(i)	Bi_2O_3 (+)3 / III AND NaBiO_3 (+)5 / V	1
1(c)(ii)	Bi_2O_3	1

14. MAR 24/P22/Q1(d)(i)

1(d)(i)	a substance that removes electrons from another / causes electron loss / gains electrons	1
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15. O/N 23/P21/Q3(a)

3(a)	compound / molecule / substance that oxidises another / is reduced	1
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16. O/N 23/P21/Q3(c)

3(c)	Cl / chlorine is both oxidised and reduced	1
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17. O/N 23/P22/Q3(b)(iii)

3(b)(iii)	P (in H_3PO_3) is (both) oxidised and reduced (simultaneously)	1
	P (is oxidised) from (+)3 / (+)III $\rightarrow$ (+)5 / (+)V AND (reduced to) -3 / -III	1

18. O/N 23/P23/Q3(a)

3(a)	compound / molecule / substance that oxidises another / is reduced	1
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19. O/N 23/P23/Q3(c)

3(c)	Cl / chlorine is both oxidised and reduced	1
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20. M/J 23/P22/Q2(e)(ii)

2(e)(ii)	$\text{NaClO} + \text{H}_2\text{O}_2 \rightarrow \text{O}_2 + \text{NaCl} + \text{H}_2\text{O}$	1
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21. M/J 23/P23/Q1(d)(i)-(ii)

1(d)(i)	$2\text{Cu}^{2+} + 4\text{I}^{-} \rightarrow 2\text{CuI} + \text{I}_2$	1
1(d)(ii)	Cu^{2+} OR CuSO_4 AND copper (species / ion) has gained / taken electron(s) from iodide (ion)	1

22. M/J 22/P22/Q1(b)(i)-(ii)

1(b)(i)	species	magnesium in Mg_3N_2	nitrogen in Mg_3N_2	1
	oxidation number	2(+) / (+)II	3- / -III	
1(b)(ii)	- redox - Mg loses electrons / Mg increases ox no / Mg is oxidised - N gains electrons / N reduces ox no / N is reduced			1

23. MAR 22/P22/Q2(a)(i)

2(a)(i)	species that donates electrons	1
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24. MAR 22/P22/Q2(c)(ii)

2(c)(ii)	(+)3 / III	1
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